

Critical Buckling Force at Conventional Sleepers

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Abstract

Conventional sleepers have historically provided a simple and economical means of initiating controlled lateral buckles in high-temperature, high-pressure subsea pipelines. However, probabilistic assessments using the critical buckling force formulations in DNV-RP-F110 [3] tend to show that these initiators are not sufficiently reliable, which pushes projects to use more expensive triggers instead, such as active sleepers. And yet, conventional sleepers, when used, have a very good track record.

In order to investigate this discrepancy, this paper investigates the forces at which lateral buckles form over conventional sleepers, using analytical formulations, linear perturbation analyses and nonlinear finite element simulations. These methods are applied to twelve representative pipeline configurations in different soil conditions.

From the insights of these assessments, the key input parameters and mechanisms are identified. This is used to propose a modification to the sleeper CBF distribution for use in probabilistic lateral buckling assessments. This approach is then applied to the case study of a pipeline in operation and shown to predict results more in line with the observed response of the pipeline.

It is expected that this improved methodology will allow the use of conventional sleepers again in projects and lead to significant cost reductions.

Nomenclature

CBF:	Critical Buckling Force
EAF:	Effective Axial Force
JIP:	Joint Industry Project
OD:	Outer Diameter
OOS:	Out-of-Straightness
PDF:	Probability Distribution Function
TDP:	Touch Down Point
UHB:	Upheaval Buckling
ZRB:	Zero-bend radius

1. Introduction

Surface-laid pipelines operating above certain temperature and pressure levels require a controlled lateral buckling design approach, in which buckle triggers are introduced at selected locations (referred to as planned buckle locations) to ensure that the resulting mechanical response remains within acceptable limits.

In the early days of controlled lateral buckling design, the buckle triggers most commonly adopted were snake-lay configurations and conventional sleepers [9, 6].

In recent years, however, the use of these relatively simple and inexpensive buckle initiation techniques has progressively declined in favour of solutions providing more robust buckle initiation mechanisms. These include ZRBs, various types of active sleepers, activated either by buoyancy or mechanical push systems, and actively imposed lateral OOS using subsea winch systems. For reel-laid pipelines, residual curvature has also been increasingly adopted [11].

This evolution in the buckle initiation methods used generally coincided with the development and adoption of probabilistic lateral buckling design approaches, first introduced through the Safebuck JIP [1, 2] and now in DNV-RP-F110 [3]. This was an important development, as the reliability of buckle triggers is a fundamental aspect of any controlled lateral buckling design, and this was not being checked very thoroughly before these methods were published.

However, when the formulations and probability distributions in DNV-RP-F110 are applied, it is often difficult to demonstrate that conventional sleepers provide a sufficiently reliable buckle initiation mechanism. This can be illustrated by comparing the distribution of critical buckling force at conventional sleepers and at unplanned buckles occurring in straight pipeline sections or route curves with typical radii, as illustrated in Figure 1 (see more details in Section 2.2).

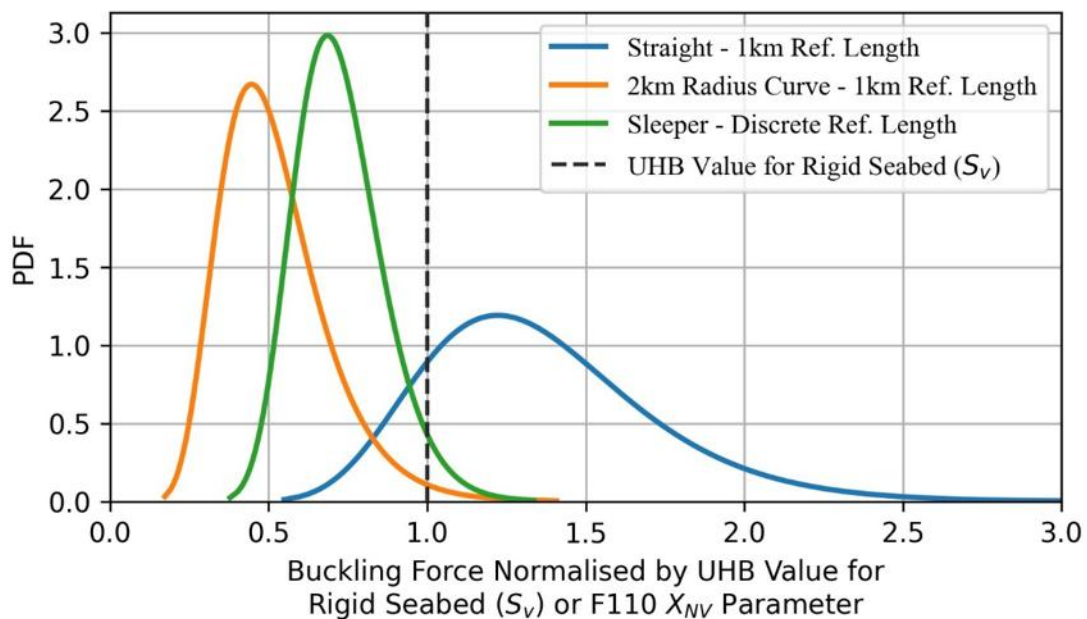


Figure 1 – PDF for the CBF at Conventional Sleeper and at Unplanned Buckles Normalised by S_v

But these predictions are in disagreement with field observations and indeed, even the operational review conducted as part of the Safebuck JIP in 2014 showed conventional sleepers to be highly reliable in triggering planned buckles.

More recently, similar observations were reported in a back-analysis of the pipelines in a deepwater development designed before the probabilistic methodologies currently in use had been fully established [12]. Although the analyses indicated a high probability of

failure to trigger at the conventional sleepers introduced by the design, operational data showed that buckles had formed at all sleeper locations.

This apparent mismatch between probabilistic predictions and observed pipeline behaviour prompted the investigation into the mechanisms responsible for buckle initiation at conventional sleepers presented in this paper. This includes an evaluation of the force at which buckles trigger over conventional sleepers and the mode shape of those buckles.

2. Operational Experience with Conventional Sleepers

2.1. Safebuck JIP

The operational experience of conventional sleepers was gathered for the Safebuck JIP and presented by Sinclair et al. in [13]. This was based on the response of 72 sleepers installed along nine pipelines with small and large diameters, within four different projects in deep and shallow waters.

A summary of the observed performance is presented in Table 1. The results show a good performance for the conventional sleepers, with 65 out of the 72 considered triggering a buckle. It is noted as well, that the 7 that did not trigger a buckle, were within two of the pipelines, which had low operating temperatures and very high residual lay tensions that were not anticipated in design. It is noted, however, that a total of 7 unplanned buckles formed on those two pipelines.

Of the 54 symmetric mode shapes, 15 were reported to have formed as anti-symmetric Mode 2 during hydrotest and evolved into symmetric modes early in operation.

Pipeline	Sleeper Height [m]	Number of Sleepers		
		Buckled with Symmetric Mode Shape ^(a)	Buckled with Anti-symmetric Mode Shape	Not Buckled
P1	1.0	4	0	0
P2	0.6	7	5	0
P3	0.6	14	0	0
P4	0.9	6 ^(b)	0	0
P5	0.9	9 ^(b)	0	0
P6 ^(c)	0.7	4 ^(b)	2	5
P7 ^(c)	0.7	4	1	2
P8	0.7	2	2	0
P9	0.9	4 ^(b)	1	0

Notes:

^(a): Modes 1 and 3 are included here given the difficulty in differentiating the two.

^(b): Some of these formed as Mode 2 during hydrotest and evolved to a symmetric mode in operation.

^(c): These pipelines had unanticipated high lay tensions (up to 50% of the driving force in operation)

Table 1 – Observed Buckling Response at Conventional Sleepers – Safebuck [13]

2.2. Experience from Flowlines in one Deep Water Field

The accuracy of the predictions made by the probabilistic lateral buckling assessment method and distributions in Appendix B of DNV-RP-F110 [3] was assessed by performing back-analyses on a series of flowlines from a field in operation and comparing the results with the observed response.

The pipelines were installed in deep waters, on a flat seabed with a soft clay surface layer, where the pipelines had significant embedment. Of these flowlines, 4 include 0.9m high conventional sleepers as buckle initiators, with a total of 9 sleepers.

Full details of the assessments and comparisons were presented in [12] and the key outcomes related to the response at the conventional sleepers are summarised in this section.

A summary of the predicted and observed response at the sleepers is summarised in Table 2. The results in the table show a significant mismatch between the very poor predicted reliability for the sleepers and the 100% success rate observed in practice. The predicted low probability of buckling at the sleepers is indeed consistent with the CBF distributions for planned and unplanned buckles illustrated in Figure 1 (which corresponds to one of these pipelines).

These results suggest that the CBF distributions for conventional sleepers in [3] overestimate the actual response; although the mismatches could also be explained by other inadequate distributions in the probabilistic assessments.

Sleeper in Route Curve	Probability of Buckling Predicted	Observed Response		
		Triggered	Mode Shape ^(a)	Berms at TDPs
Yes	5%	Yes	3	Yes
Yes	2%	Yes	3	Yes
No	20%	Yes	3	Yes
Yes	1%	Yes	3	Yes
Yes	1%	Yes	3	Yes
Yes	2%	Yes	3	Yes
No	20%	Yes	2 -> 3	Yes
No	25%	Yes	2 -> 3	Yes
No	40%	Yes	3	Yes

Notes:

^(a): Symmetric mode shapes were reported as Mode 3, although in practice it is difficult to differentiate between Modes 1 and 3.

Table 2 – Predicted and Observed Buckling at Conventional Sleepers

The observed response shows that all buckles over the sleepers ended up as Mode 3 buckles, although two of them started as Mode 2 during hydrotest and eventually evolved into Mode 3 buckles. An example of the two mode shapes is illustrated in Figure 2.

The pre and post buckling surveys also illustrate the shortening of the spans around the sleepers during operation and a certain increase in the pipe embedment at the TDPs, in agreement with the discussions in the following sections.

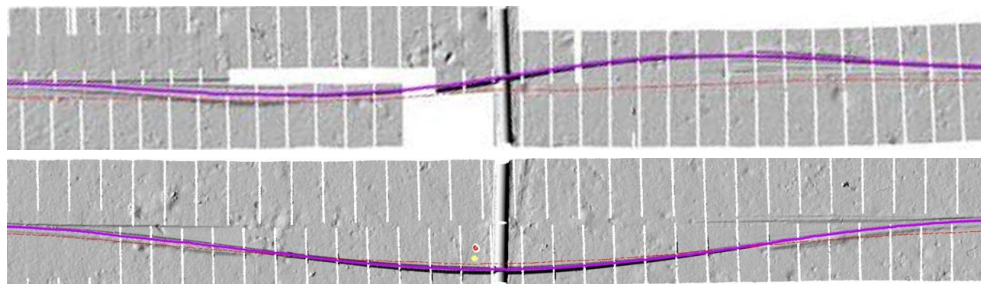


Figure 2 – Illustration of Buckling Modes 1 and 2

2.3. Overall Performance

The overall conclusion from the review of the operational performance of conventional sleepers is that they are very reliable buckle initiators, when correctly designed. This is also in line with other published data, such as Vaisberg et al. [15].

The experience also suggests that buckles tend to develop into symmetric mode shapes, although a significant number of them take anti-symmetric Mode 2 shapes.

3. Analytical Formulations

A conventional sleeper is typically considered as a way of introducing curvature in the vertical plane and thus lead to an upheaval buckling mechanism, which then rotates into a lateral buckle due to its unstable nature as it grows.

In this sense, a conventional sleeper can be seen as a prop-type feature, as considered in classic UHB analyses [7, 8, 10, 14]. Indeed, sleepers are included in the Vertical Trigger category in Appendix B of DNV-RP-F110.

But a sleeper also introduces spans either side of it and this creates the possibility of buckling due to Euler buckling mechanisms in the lateral plane, regardless of the vertical curvature introduced by the sleeper. Such modes are indeed accelerated if the pipeline contains any lateral OOS at the location of the sleeper.

In reality, both mechanisms will tend to interact in the initiation of buckling over a conventional sleeper. Both aspects are discussed in the following sections.

3.1. Classic Formulations for Prop on an Infinitely Rigid Seabed in Vertical Plane

The problem of a pipe over an unsupported prop on an infinitely rigid seabed as illustrated in Figure 3 was solved analytically in several classic papers of the 80s and 90s [7, 8, 10, 14].

In this configuration, as the effective axial compression increases, the span shortens and the reaction at the sleeper decreases (the pre-buckling flexure phase), until eventually there is lift off from the sleeper.

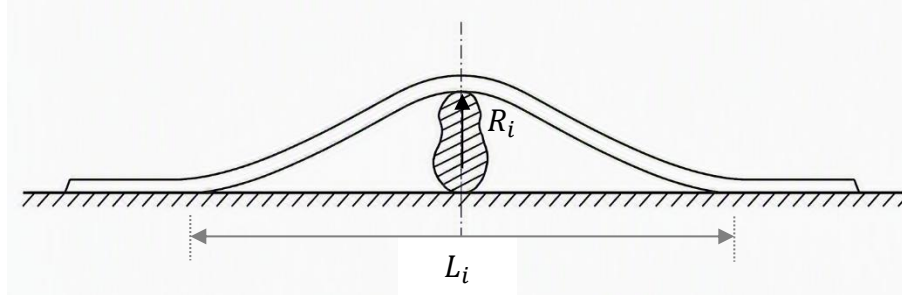


Figure 3 – Prop-Type Feature over an Infinitely Rigid Seabed Surface

The main parameters under small displacements and elastic pipe response can be calculated as follows.

- In the initial configuration, the span length and the reaction at the sleeper are:

$$L_i = 2 * \left[\frac{72 * EI * h}{SW_{oper}} \right]^{\frac{1}{4}} \quad \text{Equation 1}$$

$$R_i = \frac{2 * L_i * SW_{oper}}{3} \quad \text{Equation 2}$$

Where,

L_i : is the touch down point to touch down point initial span length

R_i : is the initial sleeper reaction

EI : is the elastic bending stiffness of the pipe

h : is the height of the sleeper

SW_{oper} : is the operating submerged weight of the pipe

- At the point of lift off from the sleeper, the span length and the compressive effective axial force are:

$$L_u = 0.775 * L_i = 2 * \left[\frac{26 * EI * h}{SW_{oper}} \right]^{\frac{1}{4}} \quad \text{Equation 3}$$

$$P_u = \frac{80.76 * EI}{L_u^2} = 3.96 * \sqrt{\frac{EI * SW_{oper}}{h}} \approx 4 * \sqrt{\frac{EI * SW_{oper}}{h}} \quad \text{Equation 4}$$

Or, as expressed in [10]:

$$\phi_w = \frac{EI * SW_{oper}}{h * P_u^2} = 0.0646 \quad \text{Equation 5}$$

Where,

L_u : is the touch down point to touch down point span length at lift off

P_u : is the compressive effective axial force at lift off

ϕ_w : is the dimensionless maximum download parameter introduced in [10]

It is noted that the basic formulation to define the critical buckling force over a conventional sleeper in Appendix B of DNV-RP-F110 (S_v) is the approximation to the force at lift off as shown in Equation 4.

In DNV-RP-F110, this equation is then multiplied by an empirically determined parameter X_{NV} , for which a lognormal distribution with a mean value of 0.725 and a standard deviation of 0.14 is proposed. This parameter is intended to capture the uncertainty in the model and the effect of the lateral OOS near the vertical imperfection.

As noted in the code, the proposed distribution for X_{NV} is based on the analysis of as-laid data from a single deep water project and care should be taken in the use of this distribution. Indeed, the code states that projects adopting vertical triggers will need to develop suitable distributions of the critical buckling force based on FE analysis models and should consider uncertainties in vertical trigger height, horizontal OOS, pipe-sleeper resistance, and pipe-soil resistance and embedment at touch down points. In practice, however, this formulation and distribution tend to be taken as presented in the code. The assessments presented in this paper address these points.

3.2. Classic Euler Buckling Formulations of Spans with Rigid Seabed

In addition to the vertical OOS, a conventional sleeper introduces a span with a length that increases with the height of sleeper as shown in Equations 1 and 3.

For such span, the point of instability can be determined from the Euler buckling equations or by solving the eigen value problem for that structural configuration.

Indeed, this does not take into account any OOS in the model (vertical or lateral), but as such, it provides an upper bound for the critical buckling force of the system.

The basic equation for the Euler buckling is presented in Equation 6.

$$P_{cr} = EI * \left(\frac{\pi}{K*L} \right)^2 \quad \text{Equation 6}$$

Where,

- P_{cr} : is the Euler buckling load
- L : is the span length considered (from TDP to TDP either side of the sleeper)
- EI : is the elastic bending stiffness of the pipe
- K : is a constant that takes the values summarised in Table 3, for the buckling mode shapes illustrated in Figure 2 and different boundary conditions.

Buckling Configuration		K
Pinned-Pinned conditions at TDPs	Mode 1	1.0
	Mode 2	0.5
Fixed-Fixed conditions at TDPs	Mode 1	0.5
	Mode 2	0.669 * 0.5

Table 3 – Euler Buckling Coefficients

In practice, the boundary conditions at the TDPs either side of the sleeper are much closer to fixed-fixed than pinned-pinned, but the coefficients of both are shown to bound the

problem. This is discussed in the next section and illustrated in the FE analyses presented in Sections 5 and 6.

Mode 1 and Mode 2 are illustrated in Figure 2. For the Mode 1 values, it is assumed that the sleeper does not provide any frictional resistance, whereas the Mode 2 does not require any assumption as the mode shape pivots on the sleeper.

If an elastic spring stiffness is introduced in the middle of the span to represent the lateral frictional resistance of the sleeper, a second term is added to Equation 6. For the more realistic fixed-fixed condition, the equation then becomes:

$$P_{cr} = EI * \left(\frac{\pi}{0.5*L} \right)^2 + \frac{2}{\pi^2} * L * K_{Sleeper} \quad \text{Equation 7}$$

Where,

$K_{Sleeper}$: is the elastic point stiffness provided by lateral frictional resistance at the sleeper

As the elastic stiffness of the sleeper increases, the buckling force increases from the Mode 1 Euler force, until the Mode 2 Euler force is reached and then the buckling mode switches from Mode 1 to Mode 2.

As mentioned above, the span length is reduced as the compressive effective axial force increases and so the value of L to consider in Equation 6 needs to be iteratively adjusted accordingly, using the equations in Section 3.1, noting however that those equations are based on an infinitely rigid seabed surface.

3.3. Modified Formulations for Spans with Elastic Seabed

The buckling loads of free spans, accounting for an elastic seabed stiffness (vertical or lateral) were studied in a series of classic papers and summarised in Fyrileiv et al. [5]. These assessments introduced the concept of the effective span length, as a way of modifying the Euler buckling loads presented in the previous section.

In these papers, the effective span lengths (vertical and lateral) are obtained either solving the non-linear equations iteratively, or by fitting the results of FE analyses. In all cases, however, the effective span length is defined as a function of $\log_{10}(\beta)$, which is defined as:

$$\beta = \frac{K_{soil} * L^4}{EI} \quad \text{Equation 8}$$

Where,

K_{Soil} : is the elastic seabed stiffness (vertical or lateral) at the span TDP

The introduction of this parameter allows obtaining the intermediate conditions between the pinned-pinned and fixed-fixed conditions in Table 3. This parameter, however, has not been derived for the case with a sleeper in the span.

3.4. Interaction of Lift-off and Euler Modes

By combining Equations 3 and 6, it is possible to determine the forces for the Euler buckling modes with the span length reduced due to the compressive force at the point of lift off (predicted by Equation 4).

By doing this, it is simple to show that the Mode 2 Euler load with the reduced span at lift off (L_u) matches exactly the lift off force (and it is 60% of the lift off force considering the initial span length L_i). Considering that at lift off any lateral resistance at the sleeper is lost, and that the Mode 1 Euler load is lower than the Mode 2 load, in this idealised case (infinitely rigid seabed, no initial imperfections and fixed-fixed boundary conditions at the TDPs), at the point of lift off, the three buckling mechanisms are predicted to occur simultaneously.

With an initial lateral imperfection at the sleeper and a small sleeper friction, however, the Mode 1 buckle could develop before the other two modes.

4. Case Studies

In order to assess the force at which pipelines buckle over a sleeper and the mode shapes they take, a series of single pipe case studies are considered. These are summarised in Table 4 and cover the full range of pipe sizes, from 6 to 32", two wall thickness values for each size based on low and high design pressure and two levels of submerged weight (light and heavy). This gives a total of 12 pipe configuration that covers the range of conditions for single pipe pipelines.

Case	OD [inch]	WT [mm]	Submerged weight [N/m]
1	6.625	4.9	45.3
2			166.2
3		11.2	201.8
4			369.7
5	12.75	9.4	171.0
6			619.2
7		21.6	751.0
8			1372.8
9	32	23.6	1090.3
10			3912.5
11		44.7	3073.3
12			7207.0

Table 4 – Case Studies Considered

The assessments presented in this paper are based on the analytical formulations detailed above and a series of perturbation and static finite element analyses presented in the following sections. These configurations are assessed with a range of seabed conditions, from medium sand to very soft clay.

The soil parameters used in the perturbation analyses are summarised in Table 5.

In the static buckling analyses, the range of soil conditions is captured by considering nominal embedments (defined as the embedment away from the TDP area) between 0 and 60% of the pipe diameter. For the lateral breakout soil friction, a range of values is assessed as shown in Figure 4. This includes low and high frictions and a case where the parameter is correlated with the nominal embedment.

In the cases where a correlation between nominal embedment and breakout lateral friction is considered, the curve shown in Figure 4 has been used. This was derived from the formulations in DNV-RP-F114, using typical properties for soft clays. It is acknowledged that this correlation is not valid in all soil conditions, but it is considered adequate to produce qualitative results.

Parameter	Soil Type	
	Medium Sand	Very Soft Clay
Vertical static stiffness, $K_{V,S}$ [kN/m/m] ^(a)	530	50
Coefficient C_L [kN/m ^{5/2}] ^(a)	12500	500
Poisson's ratio	0.35	0.50

Notes:

^(a): As per DNV-RP-F114 [4]

Table 5 – Seabed Conditions Considered in Perturbation Analyses

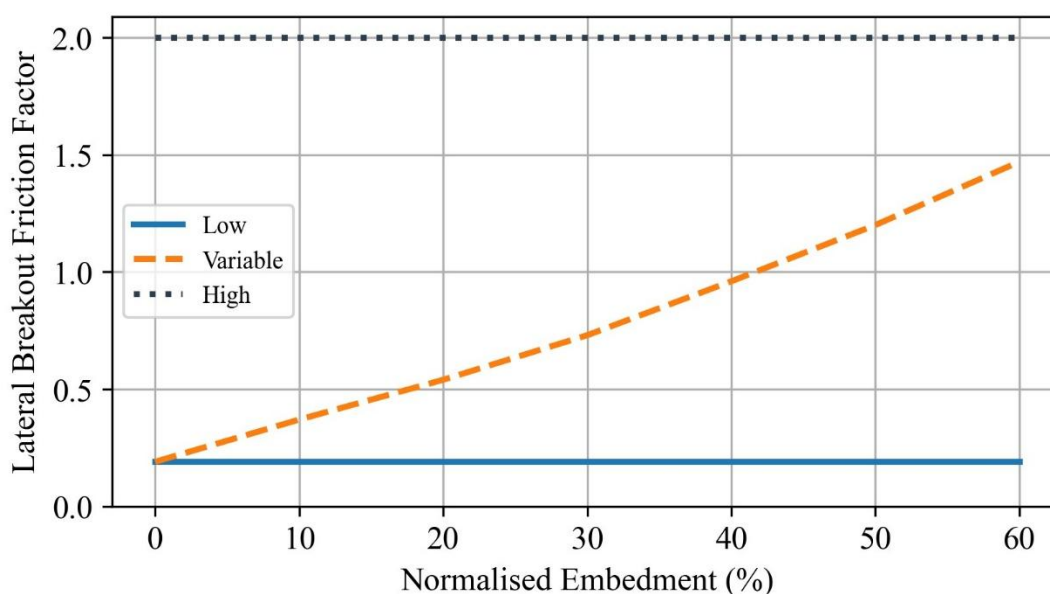


Figure 4 – Breakout Lateral Soil Friction Considered in Static Buckling Analyses

5. Eigen Value Analyses by Linear Perturbation Procedure

Eigen value analyses can be used to determine the load at which a structural system in a certain base state will become unstable (or buckle). That is done by determining the load at which the stiffness matrix of the system, linearised at the base state, becomes singular.

This is a very simple and quick type of analysis that does not have numerical convergence problems and does not require the introduction of initial imperfections. As such, perturbation analyses provide an upper bound of the critical buckling force of the actual imperfect system.

Indeed, for simple configurations, perturbation analyses match the analytical predictions of the Euler buckling loads in Section 3.2. This type of FE models allows extending those equations to more complex systems where equations may not have closed forms.

Perturbation analyses can be used to confirm the transition between Mode 1 and Mode 2 buckles, as the elastic lateral stiffness provided by the sleeper increases, matching the predictions of Equation 7.

This type of analysis can also be used for a preliminary assessment of the lateral restraint provided by the seabed beyond the span generated by the sleeper and to determine how that compares to the pinned-pinned or fixed-fixed conditions considered in Equation 6 and Table 3.

This is done by introducing an elastic stiffness equal to the lateral dynamic seabed stiffness, calculated as per Section 7.2.4 of DNV-RP-F114 [4], in the sections either side of the initial span generated by the sleeper (but of course this does not capture the reduction in span length during the pre-buckling flexure phase as the problem is linearised at the base state).

The results for the Case Studies listed in Table 4 are presented in Figure 5. The results illustrate how for these cases with an infinitely rigid seabed and no span length reduction, the buckling loads are much closer to the fixed-fixed condition than to the pinned-pinned condition, even in the cases with very soft clay (although a bit less).

This effect is assessed again in more detail in the static buckling models in Section 6.

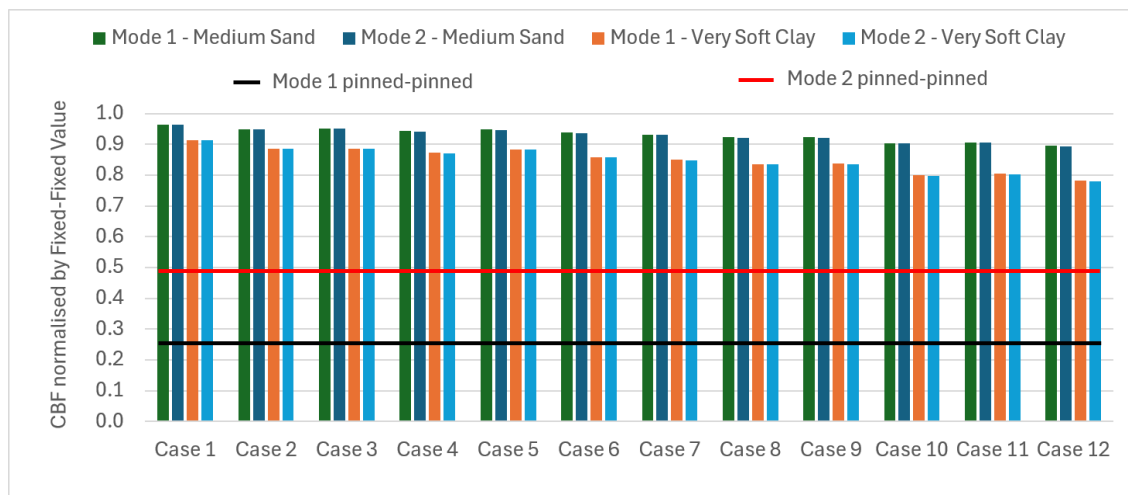


Figure 5 – Effect of Lateral Seabed Stiffness on Perturbation Buckling Loads

6. Static Buckling Analyses

The analytical formulations and perturbation analyses presented in the previous sections allow performing quick analyses, but with a number of simplifications of the problem.

In order to assess the problem in more detail, non-linear Finite Element analyses are needed. These models, however, require the introduction of geometrical imperfections to trigger buckles and so the buckling forces obtained depend on the magnitude of the imperfection introduced.

In order to determine the level of initial imperfection that is sufficient to trigger the buckle in the numerical simulation, without affecting the buckling load, sensitivity analyses are performed and comparisons are made in the configurations where a perturbation analysis can be performed. This allows determining an upper bound of the critical buckling force that does not rely on initial imperfections (that may or may not be present in the as-laid pipeline over the sleeper).

In the analyses presented in the following sections, the complexity of the models is gradually increased and the different aspects and mechanisms involved in the triggering of a buckle over a sleeper are evaluated.

6.1. 2D UHB Model on a Seabed with Vertical Stiffness

The first set of static buckling analyses consider the effect of the seabed stiffness on the 2D vertical response, the evolution of the span length (the pre-buckling flexure phase) and the force at which lift off takes place. These models do not require an initial lateral imperfection, as they consider the 2D response which includes the upset generated by the sleeper itself.

The analyses consider either a linear vertical stiffness or a bilinear vertical stiffness, with a stiffness up to a selected value of nominal embedment (defined as the embedment away from the TDP area) and a stiffness three times higher thereafter (representing the transition between the remoulded and intact seabed conditions for clay soils). Sensitivities are performed on the value of the nominal embedment condition considered, from zero (to match the infinitely rigid seabed of the analytical equations in Section 3.1) to 60% of the pipe OD. These cases are run for all pipes in Table 4.

Figure 6 and Figure 7 show the effect of the vertical seabed stiffness on the lift off force and span lengths, normalised by the values from the analytical calculations in Section 3.1 for an infinitely rigid seabed (Equation 4). As expected, the cases with the lowest nominal embedments match the analytical calculations.

The results show that although the span lengths are reduced with increasing nominal embedment, the axial force at which lift off takes place is reduced in all cases. This is because as the embedment increases, the effective height of the imperfection and the vertical curvature over the sleeper increase, as illustrated in Figure 8 for one of the case study pipes (Pipe 8). As the increase in effective height is larger in the larger diameter pipes, the trends in Figure 6 are more pronounced for the larger pipe diameters.

As also shown in Figure 6, the sensitivities with bilinear seabed stiffness show a slightly reduced effect, but the trends remain the same.

Figure 9 illustrates the evolution of the sleeper reaction with the axial compression and the effect of the nominal embedment for the same case study (Pipe 8).

These results highlight the importance of the seabed stiffness on the loss of reaction at the sleeper, and the onset of pipe lift off.

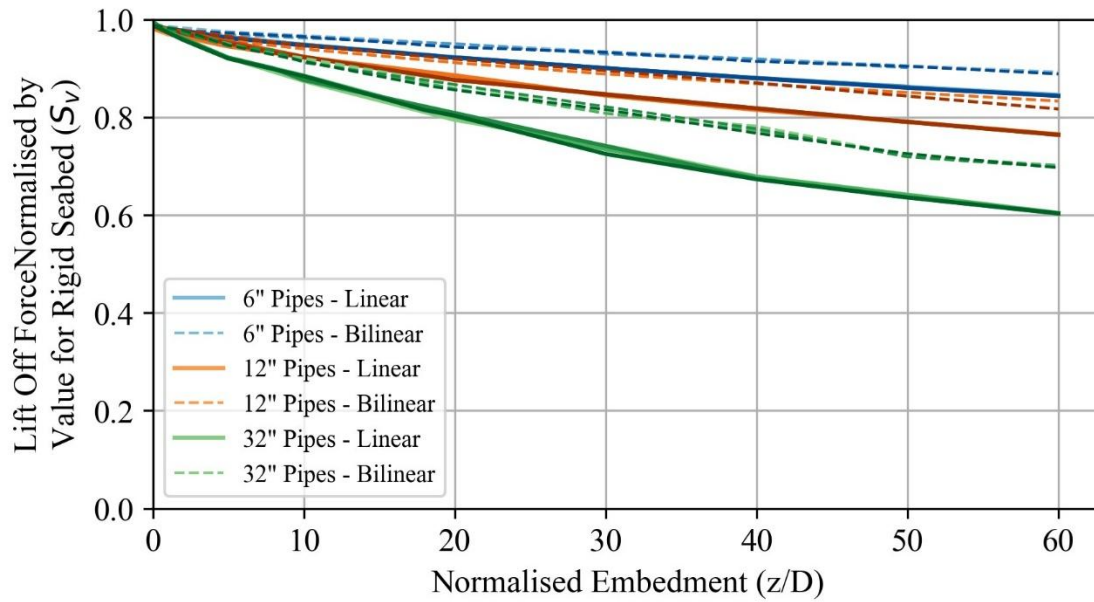


Figure 6 – Effect of Seabed Embedment on Lift Off Force Normalised by Rigid Seabed Value

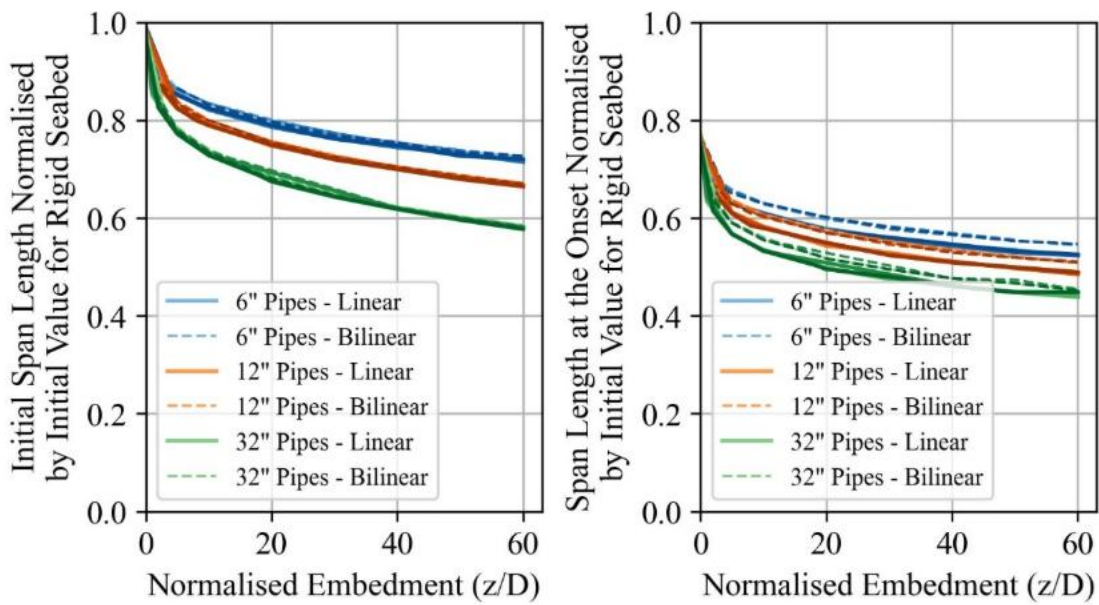


Figure 7 – Effect of Seabed Embedment on Initial and Lift Off Span Lengths Normalised by Rigid Seabed Initial Span Length

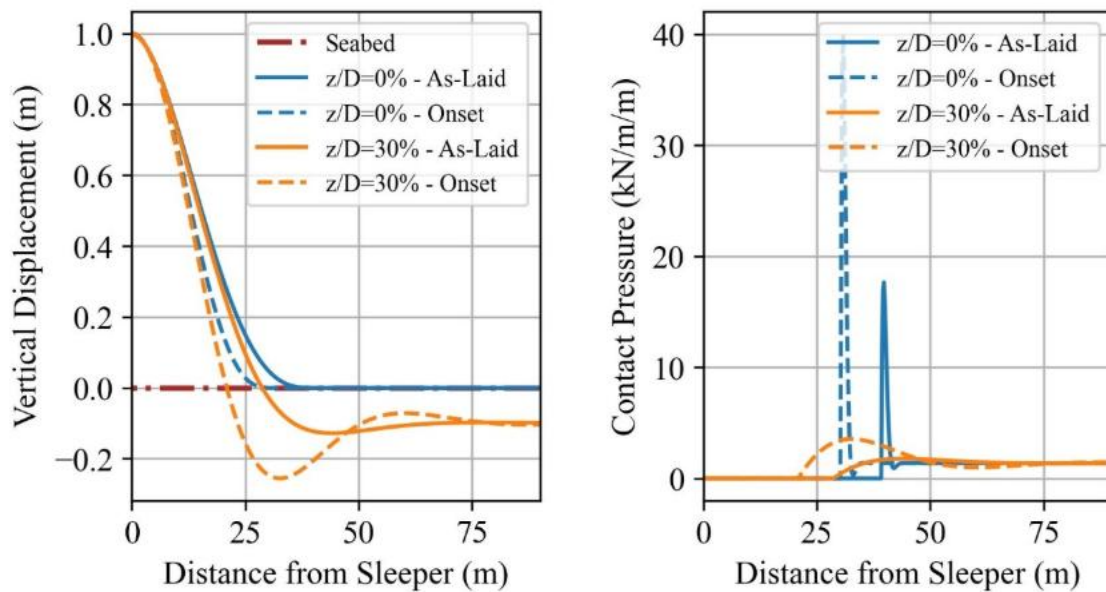


Figure 8 – Elevation and Seabed Contact Pressure Profiles – Pipe 8

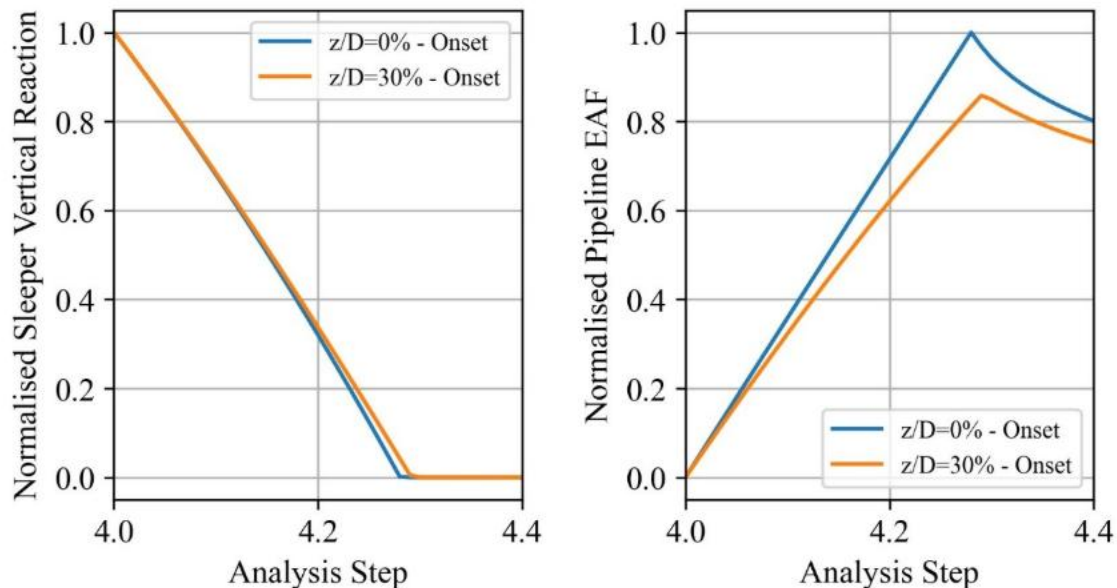


Figure 9 – Evolution of the Sleeper Vertical Reaction as the EAF Increases – Pipe 8

6.2. 3D Lateral Buckling Model with Frictionless Sleeper

The second set of static buckling analyses builds on the features of the models in the previous section and removes the 2D conditions and adds lateral imperfections to trigger lateral buckles.

The initial imperfections in these models are limited to 5cm, which is found to be sufficient to trigger the lateral buckles numerically, but sufficiently small to have a negligible

influence in the buckling forces. Imperfections with a symmetric and antisymmetric shape are considered to trigger both Mode 1 and Mode 2 buckles.

A frictionless sleeper is considered in these models to isolate the effects of the pipe-soil interaction response (vertical and lateral) on the buckling forces. This has no effect on the Mode 2 buckles and provides a lower bound for the Mode 1 buckles.

In all models, the lateral seabed friction is modelled with a contact pair, but a cap is introduced on the lateral resistance that can be developed, equal to the submerged weight times the lateral friction, i.e. additional resistance cannot be mobilised at the TDP where the contact pressures are higher than the submerged weight of the pipe.

Figure 10 and Figure 11 add the Lateral Mode 2 and Mode 1 with frictionless sleeper to the lift off results already presented in Figure 6, for the cases with linear and bilinear seabed stiffness respectively and for the three pipe sizes. In the models used to produce the results in these figures, the lateral breakout seabed friction has been correlated with the nominal embedment as per formulations in DNV-RP-F114 [4] for cohesive soils as shown in Figure 4, i.e. cases with lower nominal embedment have lower lateral friction. This is an approximation as, in practice, the correlation would have to be with the embedment at each location instead of the nominal one, but that would require a user subroutine, which has not been implemented at this stage, but the results remain qualitatively valid.

The results in these figures show that the lateral Mode 2 buckling forces are below the lift off values for the lower nominal embedments and, as the nominal embedments increase, the Mode 2 curve tends to and eventually merges with the lift off curve (and in fact the models predict lift off and mode 2 buckling triggering simultaneously). The point at which the curves merge happens at lower nominal embedments for the larger diameter pipes and for higher nominal embedments for the case with bilinear seabed stiffness (Figure 10 vs. Figure 11).

It is interesting to note that for the lowest nominal embedment cases, the Mode 2 buckling forces are considerably lower than the lift off forces (less than half). This is in disagreement with the analytical calculations in Section 3.4, which predict that the lift off and Mode 2 buckling forces should match on a rigid seabed, when span shortening is taken into account. This does not happen in these models because it is not possible to mobilise sufficient lateral seabed friction at the TDPs. In effect, the pipeline behaves as if it had an “effective lateral span length” that is longer than the actual vertical span. This is illustrated in Figure 14, where the vertical and lateral pipeline profiles at buckling onset are plotted, together with the different span lengths definitions.

This concept is equivalent to the effective span length presented in Section 3.3 for conventional spans, i.e. without the sleeper splitting the span in two. In theory, a similar exercise to the one in [5] (and the papers it references) could be performed for the case of a span with an intermediate sleeper. This could allow developing an analytical formulation of these effective span lengths, from which the buckling loads could be calculated analytically.

This mechanism is further explained in Figure 12 and Figure 13, where the sensitivity of the Mode 2 curves to the lateral breakout lateral friction and mobilisation displacements are presented respectively. As these two values are increased and reduced respectively, the

Mode 2 curve comes closer to the lift off curve, illustrating how the ‘effective lateral span’ is reduced as the lateral restraint provided at the TDPs becomes stiffer (matching the analytical calculations in Section 3.4).

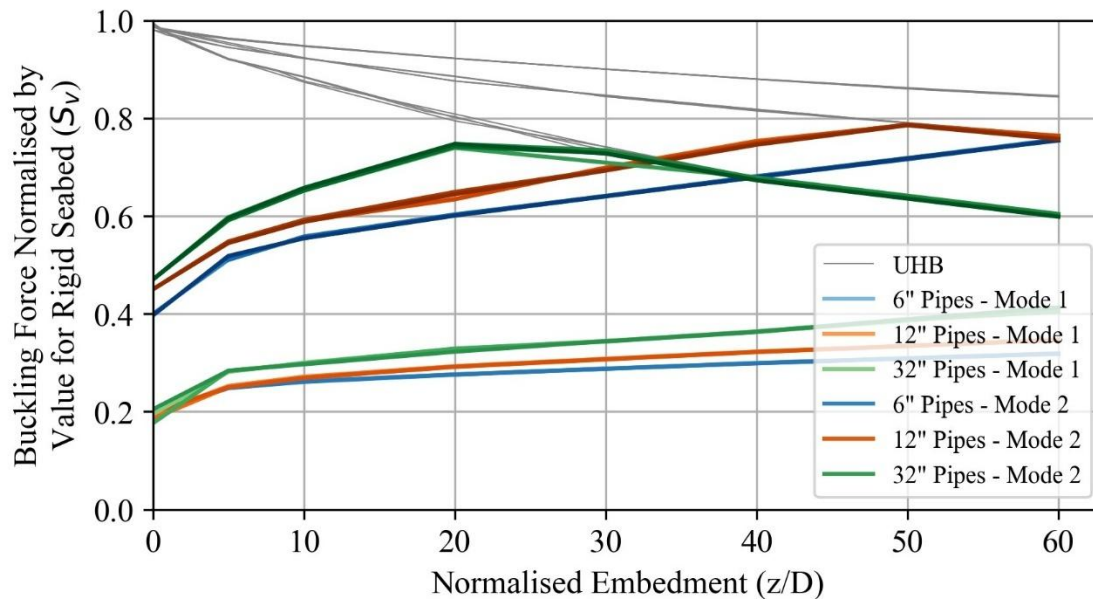


Figure 10 – Effect of Seabed Embedment on Lift Off, Lateral Modes 1 & 2 Forces Normalised by Rigid Seabed Lift Off Value (linear seabed stiffness & variable lateral friction)

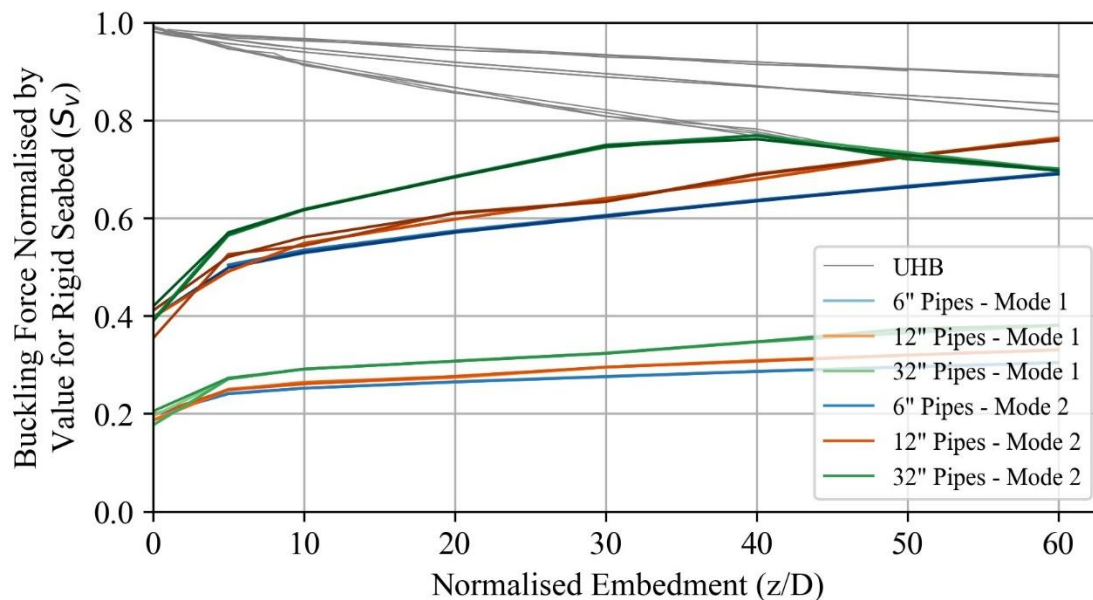


Figure 11 – Effect of Seabed Embedment on Lift Off, Lateral Modes 1 & 2 Forces Normalised by Rigid Seabed Lift Off Value (bilinear seabed stiffness & variable lateral friction)

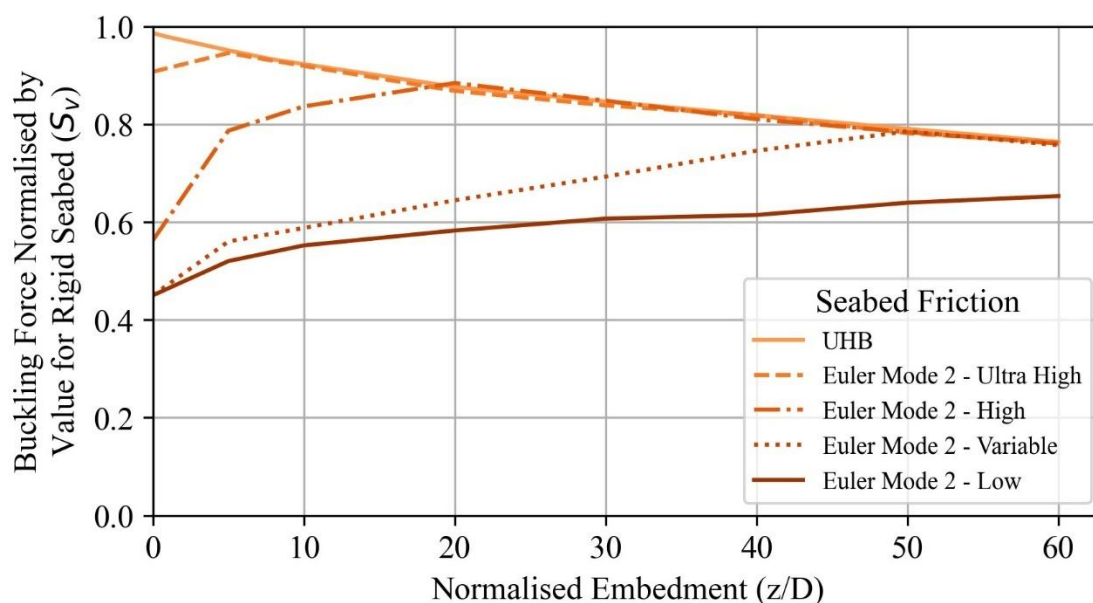


Figure 12 – Effect of Lateral Breakout Seabed Friction on Lift Off and Lateral Modes 2 Forces Normalised by Rigid Seabed Lift Off Value – Pipe 8, Linear Seabed Stiffness

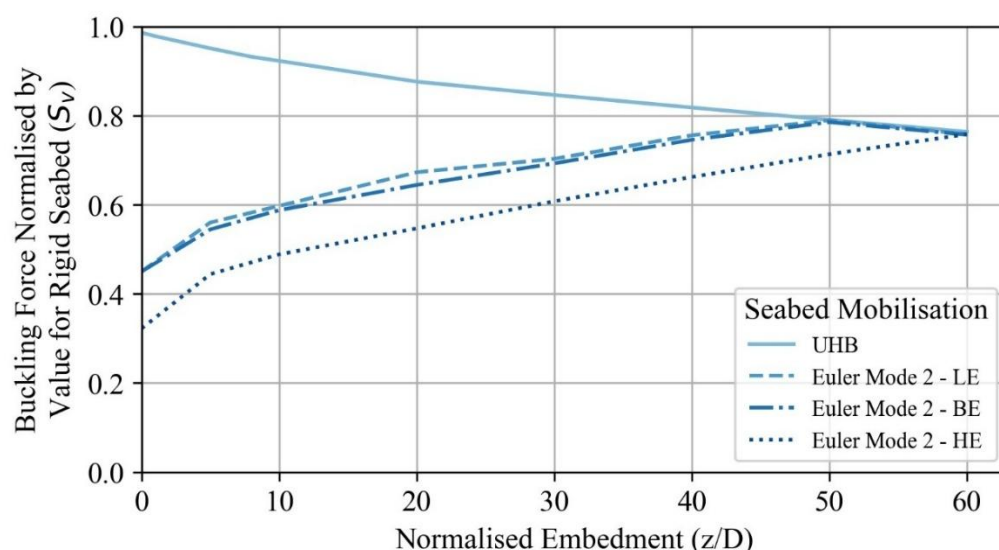


Figure 13 – Effect of Lateral Breakout Seabed Friction Mobilisation on Lift Off and Lateral Modes 2 Forces Normalised by Rigid Seabed Lift Off Value – Pipe 8, Linear Seabed Stiffness

In practice, obviously, buckling takes place in the form with the lowest buckling force. As shown in these figures, cases with low nominal embedment will buckle in Mode 2, at loads lower than the lift off force (which is used to define the CBF distribution in DNV-RP-F110 [3]). And for the cases with higher nominal embedment, the lift off force is lower than the

force with a rigid seabed used in the code. Based on this, it is clear that for all levels of embedment, buckling loads are lower than the reference formulation in DNV-RP-F110 [3]. This is discussed further in Section 7, where a modification of the formulations is proposed.

In the figures presented in this section, the Mode 1 with a frictionless sleeper is also presented for reference. The curves for this mode are obviously below the curves for the two other modes, but these are not realistic cases as sleepers cannot be frictionless. But the results are included as a lower bound for the Mode 1. In the next section, the effect of the sleeper friction in the Mode 1 buckling loads is considered.

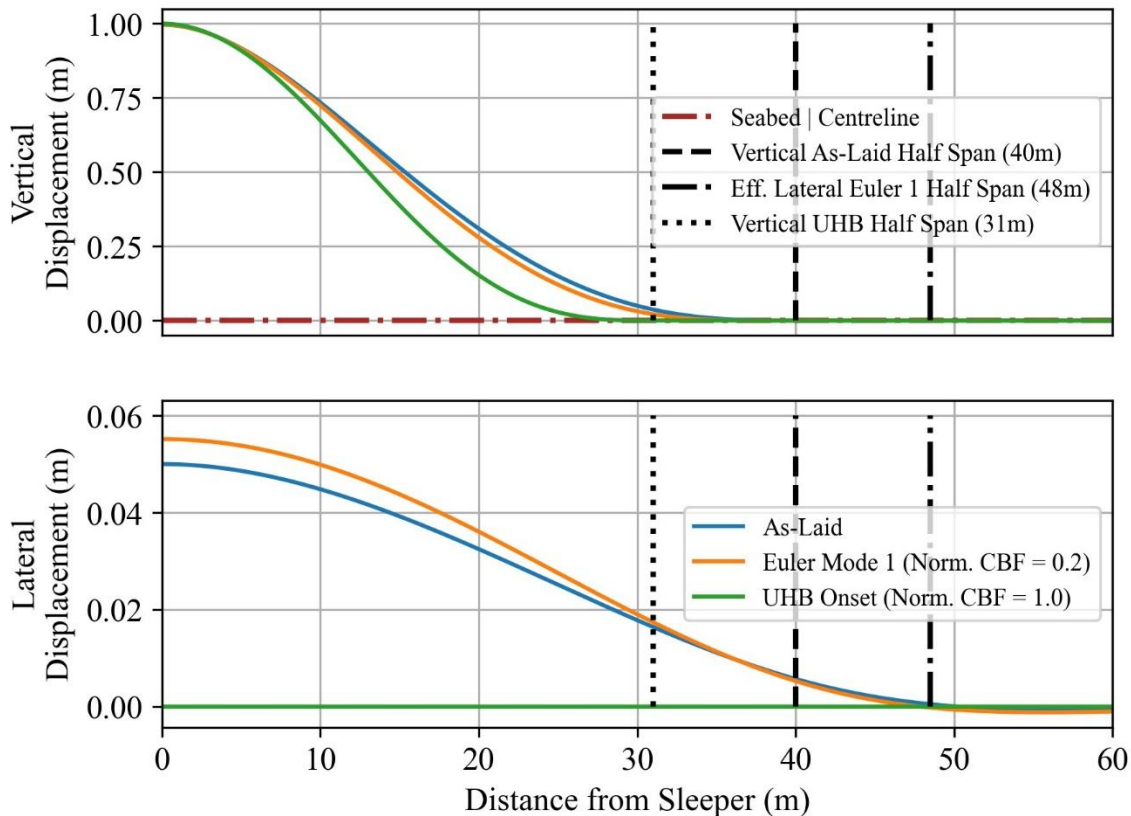


Figure 14 – Vertical and Lateral Profiles at Lift off and Mode 1 Buckling Onset Together with Vertical and Effective Lateral Span Lengths

6.3. 3D Lateral Buckling Model with Sleeper Friction

The final set of static buckling analyses adds sleeper friction to the models in the previous section and is used to assess the Mode 1 buckling loads.

In this set of analyses, instead of using the small initial imperfections only to trigger the buckles numerically, the sensitivity of the Mode 1 buckling loads to the size of the initial imperfection is assessed. This is because these types of imperfections are plausible in practice, as these can be generated during pipelay and the sleeper friction can hold them in place.

In the analyses, the sensitivity of the Mode 1 buckling force to the following parameters is considered:

- Nominal pipe embedment (as in previous sections)
- Size of the initial imperfection
- Sleeper friction
- Sleeper friction mobilisation
- Axial displacement at the buckle crown

The results of these analyses are plotted together with the curves for the loads determined in the previous sections for Lift off and Lateral Mode 2 and Mode 1 with a frictionless sleeper and presented in Figure 15. The circles in the figure correspond to the Mode 1 buckles with sleeper friction, for cases which have been generated for random combinations of nominal embedment, lateral offset and sleeper friction.

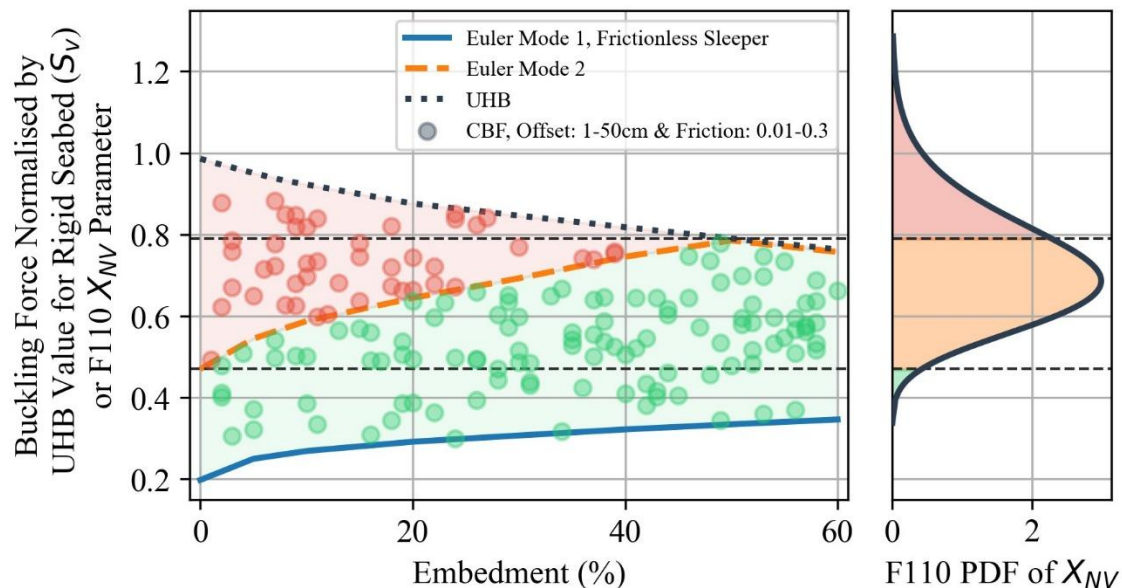


Figure 15 – Normalised Lift Off, Mode 2 and Mode 1 Buckling Forces – Pipe 8, Linear Seabed Stiffness

As shown in the figure, the new results fill the space between the curves for Mode 1 with frictionless sleeper and lift off (although some cases with large initial imperfections and low sleeper friction could potentially fall below the frictionless Mode 1 curve).

The circles that are above the Mode 2 curve are invalid, as a Mode 2 should form before the Mode 1 can develop, and have been coloured red. The points below that curve are valid and have been coloured green. The envelope of the space for the possible critical buckling forces has been shaded in green. This space should determine the distribution for the sleeper CBF. The approach proposed to consider this is presented in Section 7.

The comparison with the distribution of X_{NV} in DNV-RP-F110 is shown by plotting it on the side of the figure.

The mechanism by which these Lateral Mode 1 buckles with sleeper friction develop is as follows:

- As the axial compression increases, the span length and the vertical reaction at the sleeper reduce. This is referred to in the classic UHB papers as the pre-buckling flexure phase.
- In cases with small initial imperfections, mobilisation of the frictional resistance of the sleeper only starts when the Mode 1 force (for the span from TDP to TDP) is reached (see Figure 16). Before that, pipe deflection is mostly limited to the vertical plane.
- From this point, the frictional resistance of the sleeper is gradually mobilised.
- Lateral Mode 1 buckling with frictional sleeper takes place once the full frictional capacity of the sleeper has been mobilised. It should be noted that, as the loading progresses, the vertical reaction at the sleeper reduces and so the capacity of the sleeper for lateral frictional resistance is reduced (see Figure 16).
- In cases where the lateral Mode 2 force is reached before the sleeper friction is fully mobilised, the pipe would buckle in Mode 2 instead (although numerically this is not always captured – which explains why there are points in Figure 15 above the Mode 2 curve).

As expected, the Mode 1 buckling loads increase with sleeper friction and decrease with initial lateral imperfection. They also decrease with increasing nominal embedment, since this accelerates the unloading of the vertical reaction at the sleeper (as detailed in Section 6.1).

The load also decreases with higher sleeper friction mobilisation displacements, as this is equivalent to increasing the initial imperfection before the full sleeper frictional resistance is mobilised.

Axial displacement of the pipe over the sleeper (due to pipe expansion), in the build-up of compression, is also found to influence the buckling response (at least numerically, when axial and lateral sleeper friction are modelled as coupled). This is because part of the frictional resistance of the sleeper is used in the axial pipe direction and so is not available in the lateral pipe direction until additional lateral pipe displacement has taken place. As a result, this has the same effect as increasing the sleeper friction mobilisation displacement and so the same effect on the CBF. It is interesting to note that, at least numerically, it is possible to switch from Mode 2 to Mode 1 buckling just by adding some axial pipe displacement over the sleeper during the loading process.

The Mode 1 buckling load on a frictional sleeper can actually be determined analytically with a modified version of Equation 7 shown below, which includes the classic Euler term and a new sleeper friction resistance term:

$$P_{cr} = EI * \left(\frac{\pi}{0.5 * L} \right)^2 + \frac{\mu_{sleeper} * R_{v,sleeper} * L}{8 * \delta} \quad \text{Equation 9}$$

Where,

μ_{sleeper} : is the sleeper friction factor

$R_{V,\text{sleeper}}$: is the vertical reaction at the sleeper (which is itself a function of the axial force)

δ : is the initial lateral offset over the sleeper (plus the sleeper friction mobilisation displacement)

The L in the equation above, however, is the effective lateral span length and so to be able to use Equation 9, a solution for this parameter would have to be derived or fitted from FE analyses, as presented in [5] for the case of a simple free span. Without this, Equation 9 does not have a practical use.

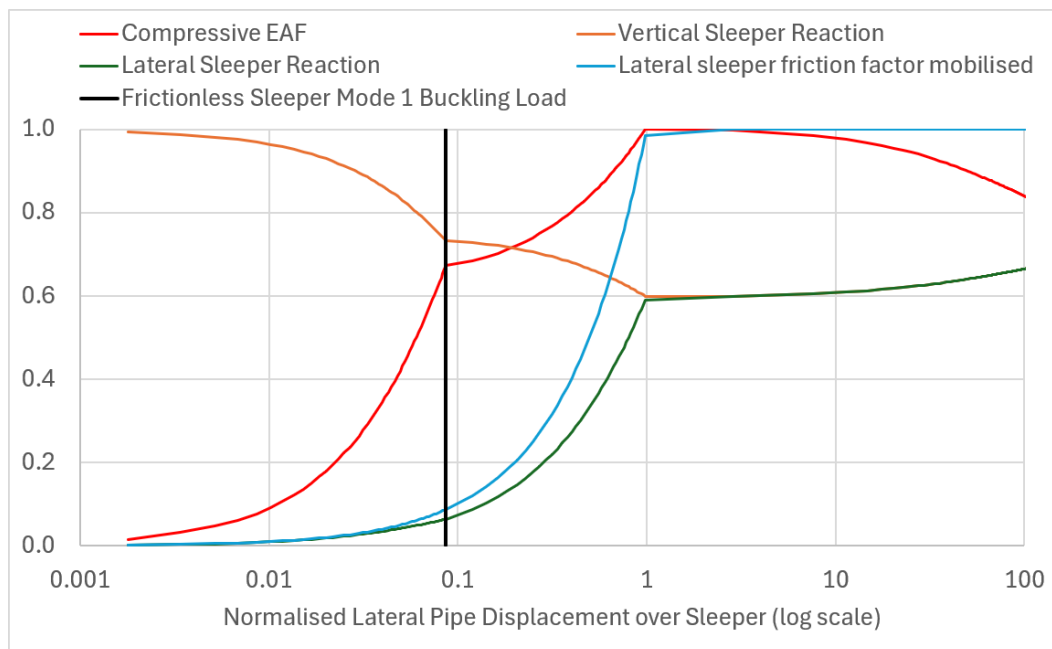


Figure 16 – Lateral Mode 1 Buckle with Sleeper Friction – Response Example¹

6.4. Overall Buckling Mechanisms

From the analyses presented in Sections 6.1 to 6.3, three mechanisms by which a pipeline can buckle over a conventional sleeper have been identified. In each as installed configuration subsea, buckling will develop following the mechanism that has the lowest load.

¹ All the quantities in the figure have been normalised.

- The lateral displacement by the pipe sleeper friction mobilisation displacement
- The compressive effective axial force by the buckling load
- The vertical reaction by the initial reaction (i.e. without any axial compression)
- The lateral sleeper reaction by the initial sleeper reaction times the sleeper friction factor
- The lateral sleeper friction mobilised by the sleeper friction factor

- Sleeper lift-off, leading to a TDP to TDP span length that is unstable as Lateral Mode 1 Euler buckling, and therefore developing as a lateral buckle (triggering at the lift off load, but being unstable as a Mode 1 frictionless sleeper case)
- Lateral Mode 2, with a shape pivoting at the point of contact with the sleeper (without lift off).
- Lateral Mode 1 with frictional resistance at the sleeper (without lift off)

The first two mechanisms are sensitive to the following parameters:

- Sleeper height, pipe bending stiffness and submerged weight, as already captured by the current code equation (Equation 4). These parameters are generally well controlled and there is limited uncertainty in their values on a given project.
- Seabed conditions at TDPs, including, vertical seabed stiffness and lateral breakout seabed friction and associated mobilisation displacement. There is clearly more uncertainty on these parameters. These parameters, however, have strong correlations and so when addressing their uncertainties, their correlations should be considered.

In addition to the above, the third mechanism is also sensitive to following parameters:

- Sleeper friction and associated mobilisation displacement
- Axial pipe displacement over the sleeper
- Initial lateral imperfection over the sleeper

It should be possible to address the uncertainty of the first two parameters, but this is much harder for the third one. This is a challenge, since the CBF for the Lateral Mode 1 with frictional sleeper is very sensitive to the size of the initial lateral imperfection (as illustrated in Figure 15).

7. Proposed Approach for CBF at Conventional Sleepers

The full space of the possible values of sleeper CBF is the green shaded area in Figure 15 (limited to the correct range of nominal embedment for each particular pipeline). A statistical treatment could potentially be done to determine these distributions, but this would require statistical data on the lateral imperfections and sufficient data to do this is not available.

Instead, the proposed approach consists of using the upper bound of this green area to define an upper bound cap for the distribution X_{NV} , as defined in DNV-RP-F110 and described in Section 3.1. Since this upper bound is defined by the envelope of the lift off and Mode 2 buckling loads, this reduces the influence of the initial lateral imperfections. It is reminded that the Mode 2 curve is obtained with minimal initial lateral imperfections and so this curve does not rely on their presence and provides an upper bound for this mode.

A set of curtailed distributions of X_{NV} is illustrated in Figure 17 for different levels of nominal embedment for the Pipe 8 case study. It is noted that these curves are presented as indicative results for this particular case and cannot be used as universal design curves.

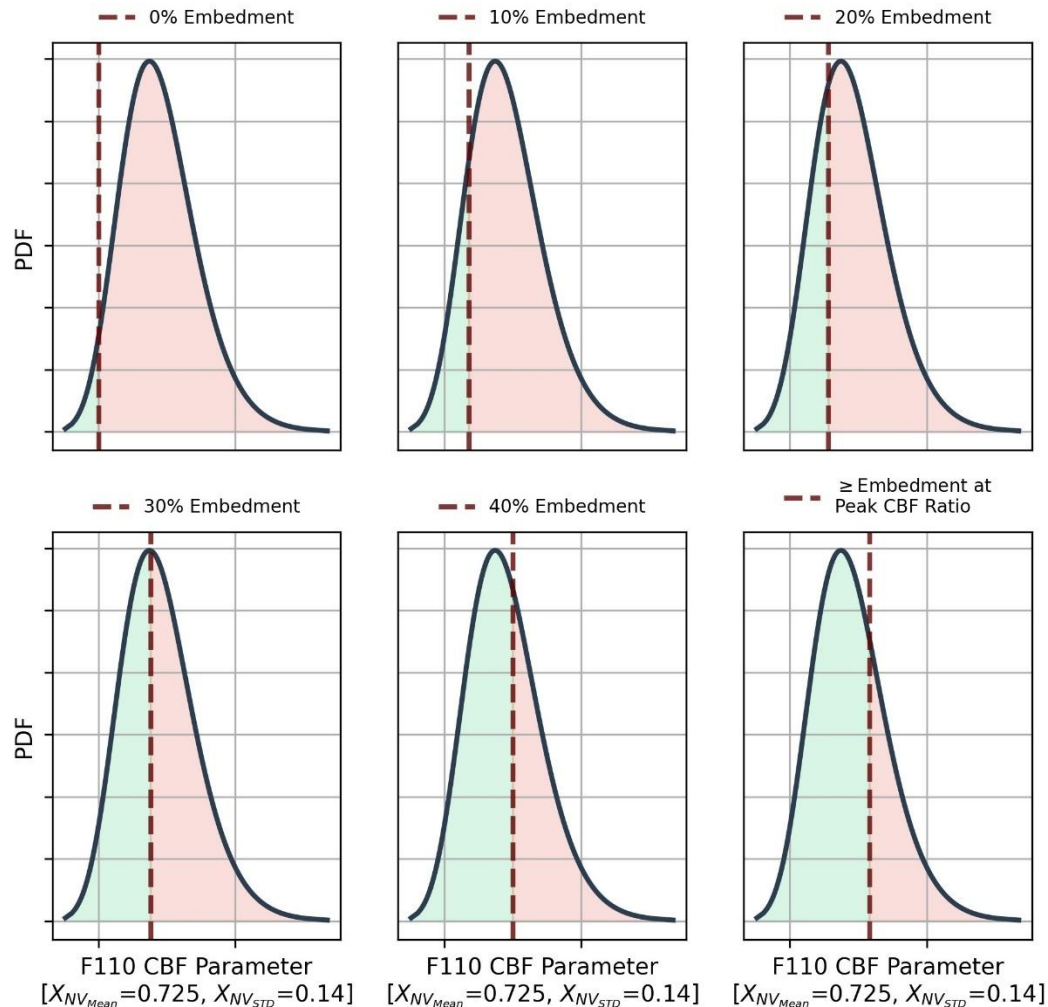


Figure 17 – Proposed Truncation of X_{NV} Distribution – Pipe 8 – Illustrative Curves

As noted in the previous section, the only aspects with significant uncertainty that influence this upper bound curve are the seabed conditions at the TDPs. At this stage, without additional geotechnical assessments and in order to produce a conservative estimate of the cut-off value of X_{NV} for a given pipeline, it is proposed to consider:

- The full range of nominal embedments
- An upper bound of the breakout lateral seabed friction (for each embedment)
- A lower bound of the breakout lateral seabed friction mobilisation displacement (for each embedment)

As illustrated in the figure, the proposed approach allows to discard a significant portion of the current distribution for all soil conditions and pipe sizes, which will significantly improve the predictions on the reliability of conventional sleepers as planned buckle triggers. The effect of this improvement on the example presented in Section 2.2 is shown in the following section.

7.1. Current Empirical Formulation in DNV-RP-F110

As summarised at the end of Section 3.1, the distribution of X_{NV} that is proposed in Appendix B of DNV-RP-F110 [3] to factor the classic lift off force (Equation 4) was developed empirically using FE analyses comparable to the ones presented in Section 6.3 of this paper, using survey data from as-laid pipelines over sleepers.

And so, it is reasonable to question why these analyses did not reach the same conclusions as this paper. The answer to this question is illustrated in Figure 15, which shows how when symmetric imperfections are introduced in the FE models, the models do not capture the Mode 2 buckles and form at higher loads. This is considered a numerical feature, given that Mode 2 buckles at lower loads can be triggered numerically introducing very small antisymmetric initial imperfections (see Section 6.2). To some extent, this is also validated by some of the eigen value linear Perturbation analyses.

8. Impact in Probabilistic Buckle Formation Assessments

The approach proposed in the previous section is used to repeat the probabilistic lateral buckling back-analyses for two of the pipelines presented in Section 2.2. The CBF distributions used are shown in Figure 18 and the probabilities of buckling obtained are presented in Table 6, together with the original predictions and the observed response already shown in Table 2.

The results suggest a closer alignment between the predictions and the observed response.

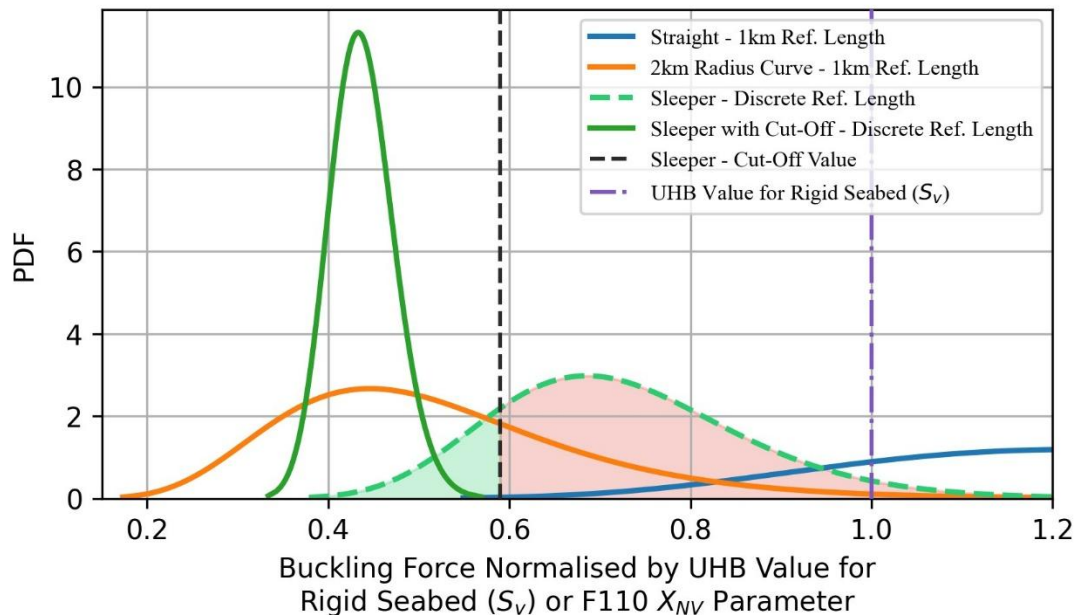


Figure 18 – PDF for the CBF at Conventional Sleeper (F110 and Proposed) and at Unplanned Buckles, Normalised by S_v [Table 6, 1st and 2nd Pipelines]²

² The distribution Sleeper with Cut-Off has been reconstructed from the curtailed original distribution

Sleeper in Route Curve	Probability of Buckling ^(a)		Observed Response		
	F110	Proposed	Triggered	Mode Shape ^(b)	Berms at TDPs
Yes	5%	57%	Yes	3	Yes
Yes	2%	50%	Yes	3	Yes
No	40%	96%	Yes	3	Yes

Notes:

^(a): Probabilities predicted either with the CBF distribution at the sleepers from DNV-RP-F110 [3] or as proposed in this paper

^(b): Symmetric mode shapes were reported as Mode 3, although in practice it is difficult to differentiate between Modes 1 and 3.

Table 6 – Predicted (F110 and Proposed) and Observed Buckling at Conventional Sleepers

9. Conclusions and Recommendations

This paper has reviewed and explained in detail the mechanisms by which pipelines form lateral buckles over conventional sleepers. This work was motivated by the apparent mismatch between the poor reliability predicted for these initiators using current design guidelines and their successful operational experience.

From the results of these assessments, a new approach to determine the distribution of critical buckling force at conventional sleepers for use in probabilistic lateral buckling assessments has been developed and presented in this paper.

The approach proposed is based on introducing an upper bound cap to the distributions of X_{NV} in DNV-RP-F110 [3]. This cap is derived from FE analyses that consider Lift off and Lateral Mode 2 buckles, without any reliance on initial lateral imperfections over the sleeper.

It is noted that the formulation currently in DNV-RP-F110 for the sleeper CBF is independent of the seabed conditions, which are illustrated in the paper to have significant influence on the buckling load, and which are captured in the proposed cap.

For the example pipeline considered in this paper, this approach produces probabilities of buckling at the conventional sleepers more in line with the observed response.

It is expected that the methodology and results presented in this paper will encourage a return to the use of conventional sleepers instead of the active sleepers more common nowadays and that this will allow reducing project costs.

The proposed approach has outlined how to address the influence of uncertain parameters, but some improvements could still be achieved. To address this, the following future work is recommended:

- Develop a methodology to select the range of nominal embedments considered and the non-linear vertical seabed stiffness.
- Develop a methodology to correlate the lateral breakout seabed friction with the actual pipe embedment along the TDP area (this will need to be implemented in a user subroutine).

Finally, this paper has presented the three mechanisms by which a pipeline can laterally buckle over the sleeper (at least numerically) and provided methods to estimate the loads at which each of these mechanisms trigger.

The operational experience provides the mode shape in which buckles form in practice. But no data is available on how these buckles developed and therefore by which mechanism they buckled. This could be addressed by visually recording the development of such buckles. It would be interesting to do that at sleepers installed within route curves and at sleepers where the pipeline has been installed with minimal lateral OOS. This may allow to fine tune and improve the predictions for the different configurations.

In particular, the limited operational evidence presented in Table 2 suggest that Mode 2 buckles are more likely to develop in sections of nominally straight pipe and not in route curves. This would be consistent with pipelines laid over conventional sleepers within route curves being more prone to contain initial lateral imperfections, which encourage the development of the lateral buckles as mode 1 with frictional sleeper resistance. By investigating this, it may be possible to improve the predictions for these two scenarios.

10. References

1. Bruton, D., Carr, M. SAFEBUCK JIP – Story so Far: Safe Design of Hot On-Bottom Pipelines with Lateral Buckling. HPHT Conference, June 2008.
2. Cosham, A., Carr, M., Bruton, D., Leslie, D. The Reliable Initiation of Lateral Buckles – SAFEBUCK JIP. SUT Annual Conference, February 2009.
3. DNV-RP-F110, Global Buckling of Submarine Pipelines, September 2021.
4. DNV-RP-F114, Pipe-Soil Interaction for Submarine Pipelines, May 2021
5. Fyrileiv, O., Mørk, K. Structural Response of Pipeline Free Spans Based on Beam Theory, 21st International Conference on Offshore Mechanics and Arctic Engineering, OMAE2002-28458, Oslo, Norway, 2002
6. Harrison, G.E., Brunner, M.S, Bruton, D.A.S. King Flowlines – Thermal Expansion Design and Implementation, Proceedings of the Annual Offshore Technology Conference. OTC 15310, 2003.
7. Hobbs, R.E. In-service Buckling of Heated Pipelines, Journal of Transportation Engineering, ASCE, Vol. 110, No. 2, March 1984.
8. Ju, G.T., Kyriakides, S. Thermal Buckling of Offshore Pipelines, Journal of Offshore Mechanics and Arctic Engineering, Vol. 110, 1988.
9. Matheson, I., Carr, M., Peek, R., Sanders, P. and George, N. Penguins Flowline Lateral Buckle Formation Analysis and Verification. 23rd International Conference on Offshore Mechanics and Arctic Engineering. Vancouver, OMAE 2004.
10. Palmer, A.C., Ellinas, C.P., Richards, D.M., Guijt, J. Design of Submarine Pipelines Against Upheaval Buckling, Offshore Technology Conference, OTC 6335, 1990
11. Roy, A., Rao, V., Charnaux, C., Ragupathy, P., Sriskandarajah, T. Straightener Settings for Under-Straight Residual Curvature of Reel Laid Pipe, 33rd International Conference on Ocean, Offshore and Arctic Engineering, OMAE2014-24513, San Francisco, California, USA, 2014.

12. Sicilia, C., Ripoll, I. Retrospective Analysis of the Lateral OOS and Buckling Response of Operational Deep-Water Pipelines, ISOPE Conference, June 2024.
13. Sinclair, F., Carr, M., Bruton, D., Farrant, T. Design Challenges and Experience with Controlled Lateral Buckle Initiation Methods, 28th International Conference on Offshore Mechanics and Arctic Engineering, OMAE2009-79434, Honolulu, Hawaii, USA, 2009
14. Taylor, N. Tran V. Prop-Imperfection Submarine Pipeline Buckling, Marine Structures 6, 1993
15. Vaisberg, O., Solano, R., Duret, H., Jebali, A., Scultori, M., Ochi, V. Lateral Buckling Re-Assessment after As-built Survey of Mero 1 Project Deepwater Flowlines, ISOPE Conference, June 2024.